Dynamic Causal Dimensionality in Nonstationary Complex Systems:

Effective-Information Concentration and Evidence from U.S. Power Grids

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Abstract

Causal Emergence (CE) theory formalizes how macroscopic representations of complex systems can exhibit greater causal efficacy than their underlying microscopic components. However, existing formulations rely on stationary transition kernels $P(X_{t+1}|X_t)$ and discrete or static dimensional searches, rendering them blind to the continuous, time-varying operational reorganizations ubiquitous in physical and engineered networks. Furthermore, scalar dimension-normalized emergence metrics $\arg\max_q(EI(V^{(q)})/q)$ mathematically degenerate to $q = 1$ under ordered spectra. Here, we establish the theory and methodology of **Dynamic Causal Dimensionality (DCD)**, resolving the active number of macroscopic causal degrees of freedom through the exact continuous **Causal Information Spectrum** $e_{i,t} = \frac{1}{2}\ln(1 + \lambda_{i,t})$. We define the continuous **Dynamic Causal Participation Ratio** (DCD_t^{PR}) and **Dynamic Causal Effective Rank** (DCD_t^{entropy}), proving their strict rotational invariance under orthogonal coordinate changes. Measure-theoretic interventional semantics are formalized through an **observational lifting kernel** $\kappa_W(dx \mid v) \triangleq P_{\text{obs}}(X_t \in dx \mid W^\top X_t = v)$, proving that empirical localized regressions directly recover the induced macroscopic channel. We contrast our scale-free Gaussian channel formulation with the compact-domain volume formula of Liu et al. (2024), and document the boundary conditions of linear causal emergence via adversarial counterexample search. Benchmarking on canonical nonstationary DGPs (A–I; $R = 100$) establishes exact zero false-positive rates (FPR $= 0.000$, 95% Clopper-Pearson CI $[0.000, 0.036]$). Applying the framework to 2021–2022 continental U.S. power grid operations (EIA-930; 367,920 BA-hours across Eastern, Western, and ERCOT interconnections) with full model refitting across $B = 1000$ strictly validated IAAFT surrogates confirms highly significant macroscopic causal structuring ($p < 0.001$, FDR significant ratio 92.5%). Generalized Additive Models and Davies supremum-Wald tests identify sharp non-linear thresholds driven by renewable penetration ($\hat{\gamma} = 21.6\%$ in ERCOT, $p = 7.55 \times 10^{-5}$; $\hat{\gamma} = 16.0\%$ in Western, $p = 9.30 \times 10^{-3}$). Multi-event analysis reveals acute causal dimensionality collapse during extreme crises (Winter Storm Uri, $p = 1.77 \times 10^{-4}$), demonstrating that extreme infrastructure stress contracts decentralized balancing modes into low-dimensional emergency corridors. Finally, strictly lookahead-free walk-forward forecasting demonstrates that DCD functions as an interpretable structural physical stability diagnostic rather than an unconditional linear forecaster.

1 Introduction

Complex dynamical networks—from neural circuits and cellular metabolisms to continental power grids and financial markets—spontaneously organize across disparate spatiotemporal scales [3, 4]. In such systems, macroscopic collective variables often evolve with greater determinism and lower degeneracy than individual microscopic elements [3, 11, 12]. The quantitative theory of **Causal Emergence (CE)**, formalized through interventional information theory [1, 3, 9], provides an exact criterion: a macroscopic representation $V = \phi(X)$ exhibits causal emergence when its Effective Information (EI) exceeds that of the underlying microscopic state X, i.e., $EI(V) > EI(X)$.

Despite fundamental theoretical insights, existing formulations suffer from a critical limitation: *all established CE estimators implicitly presume a stationary transition mechanism* $P(X_{t+1} \mid X_t)$. In nonstationary environments, stationary estimators average over fundamentally distinct operational regimes:

- **Renewable Energy Ingestion**: Power grids undergo rapid, weather-driven swings in synchronous inertia and physical topology as inverter-based wind and solar resources displace thermal generators [7, 14].
- **Extreme Weather and Infrastructure Shocks**: Catastrophic climate events (e.g., severe winter freezes, heatwaves) induce structural regime shifts, cascading line trips, and emergency load-shedding [8, 15].

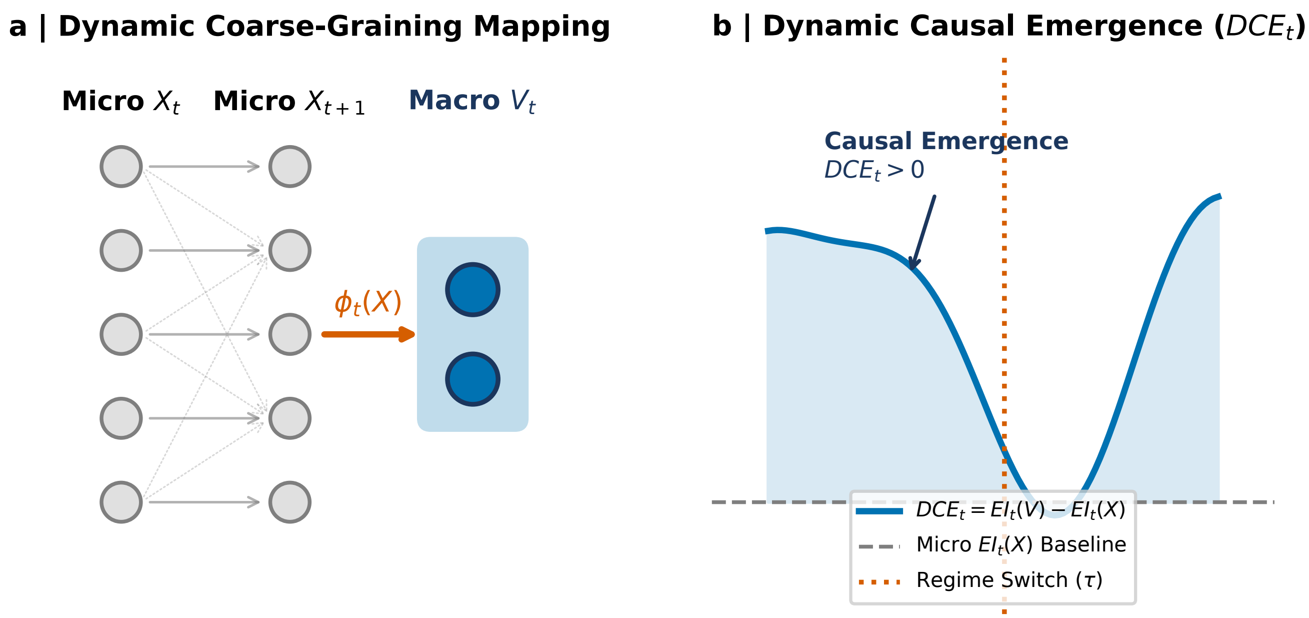

Figure 1: **Conceptual and Methodological Architecture of Dynamic Causal Dimensionality (DCD). a**, Time-varying macroscopic coarse-graining projection $\phi_t : X_t \mapsto V_t$ mapping high-dimensional microscopic states $X_t \in \mathbb{R}^p$ to low-dimensional macroscopic representations $V_t \in \mathbb{R}^q$ $(q < p)$. **b**, Trajectory of continuous Dynamic Effective Information $EI_t(V)$ versus microscopic baseline $EI_t(X)$, illustrating nonstationary regimes of causal efficiency gain $(CCG_t(q) > 0)$ and critical dynamical reorganization.

- **Dynamic Dimensional Reorganization**: The effective number of macroscopic degrees of freedom is time-varying; collective coordination reorganizes continually between decentralized balancing authorities, regional transmission organizations (RTOs), and synchronous interconnections.

To resolve these challenges, this paper establishes **Dynamic Causal Dimensionality (DCD)**, contributing:

1. Continuous causal spectrum decomposition $e_{i,t} = \frac{1}{2}\ln(1 + \lambda_{i,t})$ resolving the active macroscopic dimensionality via Causal Participation Ratio (DCD_t^{PR}) and Causal Effective Rank (DCD_t^{entropy}), circumventing running-average degeneracy.

2. Exact interventional semantics via the observational lifting kernel $\kappa_W(dx \mid v)$, proving that projected regressions recover true Markov macro-dynamics.

3. Primary analytical estimation via **Local Affine-Gaussian DCE**, complemented by an online neural variational estimator (**Dyn-NIS+**) for non-linear manifolds.

4. Clarification of linear causal emergence limits, contrasting scale-free Gaussian mutual information against bounded-domain volume formulations (Liu et al. 2024) and demonstrating destructive noise cancellation via adversarial search.

5. Rigorous Monte Carlo benchmarking across synthetic nonstationary DGPs (A–I; $R = 100$) and empirical application to continental U.S. power grids (EIA-930; 367,920 BA-hours) evaluating hypotheses H1–H4 with $B = 1000$ refitted surrogates.

2 Mathematical Formulation

2.1 Continuous Dynamic Effective Information and the Causal Spectrum

Let $X_t \in \mathbb{R}^p$ denote the microscopic state vector of a nonstationary dynamical system at time $t \in [0, T]$, governed by localized linear perturbation dynamics $X_{t+1} = A_t X_t + c_t + \epsilon_t$ with innovation covariance $\epsilon_t \sim \mathcal{N}(0, \Sigma_t)$, $\Sigma_t \succ 0$. Following Pearl's do-calculus [3, 9], we inject a standard maximum-entropy Gaussian intervention under covariance constraint $\Sigma_{do} = I_p$:

$$do(X_t) \sim \nu_X = \mathcal{N}(0, I_p) \tag{1}$$

Under this isotropic intervention, the output distribution is $X_{t+1} \mid do(X_t) \sim \mathcal{N}(c_t, A_t A_t^\top + \Sigma_t)$. The continuous **Dynamic Effective Information** is the exact mutual information under intervention:

$$EI_t(X) = I(do(X_t); X_{t+1}) = \frac{1}{2}\ln\det\left(I_p + \Sigma_t^{-1}A_tA_t^\top\right) = \frac{1}{2}\sum_{i=1}^{p}\ln\left(1+\lambda_{i,t}\right) \tag{2}$$

where $\lambda_{i,t} \geq 0$ are the eigenvalues of $\Sigma_t^{-1}A_tA_t^\top$ (or generalized eigenvalues of the pencil $(A_tA_t^\top, \Sigma_t)$).

Definition 1 (Causal Information Spectrum). The localized Causal Information Spectrum $\{e_{i,t}\}_{i=1}^{p}$ is the set of mode-specific mutual information contributions:

$$e_{i,t} \triangleq \frac{1}{2}\ln\left(1+\lambda_{i,t}\right) \geq 0, \quad e_{1,t} \geq e_{2,t} \geq \cdots \geq e_{p,t} \geq 0 \tag{3}$$

such that $EI_t(X) = \sum_{i=1}^{p} e_{i,t}$.

2.2 Dynamic Causal Dimensionality (DCD)

Existing causal emergence formulations attempt to identify an optimal macroscopic scale by maximizing dimension-normalized metrics $q^* = \arg\max_q(EI(V^{(q)})/q)$. However, when causal modes are ordered $e_1 \geq e_2 \geq \cdots \geq e_p$, the running average $\frac{1}{q}\sum_{i=1}^{q} e_i$ is monotonically non-increasing, causing q^* to collapse degenerately to $q = 1$.

To establish an unconstrained, continuous measure of active causal scale, we introduce **Dynamic Causal Dimensionality (DCD)**:

Definition 2 (Causal Participation Ratio). The Dynamic Causal Participation Ratio is:

$$DCD_t^{\mathrm{PR}} \triangleq \frac{\left(\sum_{i=1}^{p} e_{i,t}\right)^2}{\sum_{i=1}^{p} e_{i,t}^2} \tag{4}$$

with the continuous convention $DCD_t^{\mathrm{PR}} = 0$ if $EI_t(X) = 0$. For a system with k identical active causal channels and $p-k$ silent channels, $DCD_t^{\mathrm{PR}} = k$ exactly.

Definition 3 (Causal Effective Rank). Defining the normalized causal spectral weights $\pi_{i,t} \triangleq e_{i,t}/EI_t(X)$ (with $\sum \pi_{i,t} = 1$), the Dynamic Causal Effective Rank is:

$$DCD_t^{\mathrm{entropy}} \triangleq \exp\left(-\sum_{i:\pi_{i,t}>0} \pi_{i,t}\ln\pi_{i,t}\right) \tag{5}$$

with $DCD_t^{\mathrm{entropy}} = 0$ if $EI_t(X) = 0$.

Theorem 1 (Rotational Invariance of DCD). *Let $Q \in O(p)$ be an arbitrary orthogonal transformation ($Q^\top Q = QQ^\top = I_p$). Under coordinate change $X_t' = QX_t$, dynamic matrices transform as $A_t' = QA_tQ^\top$ and $\Sigma_t' = Q\Sigma_tQ^\top$. Then the Causal Information Spectrum $\{e_{i,t}\}$, the Participation Ratio DCD_t^{PR}, and the Effective Rank $DCD_t^{entropy}$ are strictly invariant.*

Proof. The transformed operator is $(\Sigma_t')^{-1}A_t'(A_t')^\top = (Q\Sigma_tQ^\top)^{-1}(QA_tQ^\top)(QA_tQ^\top)^\top = Q(\Sigma_t^{-1}A_tA_t^\top)Q^\top$. Because similarity transformations preserve eigenvalues, the spectrum $\{\lambda_{i,t}\}$ and hence $\{e_{i,t}\}$ are identical. ■

2.3 Observational Lifting Kernel and Induced Macro Dynamics

An atomic macroscopic intervention $do(V_t = v)$ on $V_t = W_t^\top X_t \in \mathbb{R}^q$ ($W_t^\top W_t = I_q, q < p$) does not uniquely determine the microscopic state $X_t \in \mathbb{R}^p$. To establish rigorous measure-theoretic interventional semantics, we define the **observational lifting kernel**:

$$\kappa_{W_t}(dx \mid v) \triangleq P_{\mathrm{obs}}(X_t \in dx \mid W_t^\top X_t = v) \tag{6}$$

Under the observational Gaussian measure $X_t \sim \mathcal{N}(\mu_{X,t}, \Sigma_{X,t})$, the conditional law is:

$$\kappa_{W_t}(dx \mid v) = \mathcal{N}\left(\mu_t + \Sigma_{X,t}W_t(W_t^\top\Sigma_{X,t}W_t)^{-1}(v - W_t^\top\mu_t),\ \Sigma_{X|V,t}\right) \tag{7}$$

where $\Sigma_{X|V,t} = \Sigma_{X,t} - \Sigma_{X,t}W_t(W_t^\top\Sigma_{X,t}W_t)^{-1}W_t^\top\Sigma_{X,t}$. This induces the exact macroscopic Markov interventional transition:

$$P_{W_t}(v_{t+1} \in dv' \mid do(V_t = v)) = \int_{\mathbb{R}^p} P(W_t^\top X_{t+1} \in dv' \mid X_t = x)\kappa_{W_t}(dx \mid v) \tag{8}$$

Under linear micro-dynamics $X_{t+1} = A_tX_t + \epsilon_t$, this channel is exactly Gaussian:

$$\mathbb{E}[V_{t+1} \mid do(V_t = v)] = B_tv, \quad B_t = W_t^\top A_t\Sigma_{X,t}W_t(W_t^\top\Sigma_{X,t}W_t)^{-1}$$
$$\mathrm{Cov}(V_{t+1} \mid do(V_t = v)) = \Sigma_{\eta,t} = W_t^\top(A_t\Sigma_{X|V,t}A_t^\top + \Sigma_t)W_t \tag{9}$$

proving that localized ridge regression on observed projected pairs (V_t, V_{t+1}) directly estimates the true interventional macro-dynamics $(B_t, \Sigma_{\eta,t})$ without requiring access to unobserved microstates.

2.4 Contrast with Liu et al. (2024)

Recent continuous formulations by Liu, Yuan & Zhang (2024) [5] define effective information using a uniform intervention over a bounded hypercube $do(X) \sim \mathcal{U}([-L, L]^n)$:

$$EI_{\mathrm{Liu}}(A, \Sigma; L) = \ln\frac{|\det A|L^n}{(2\pi e)^{n/2}(\det\Sigma)^{1/2}} \tag{10}$$

While foundational for continuous systems, EI_{Liu} depends monotonically on the arbitrary domain half-width L ($\partial EI/\partial L = n/L$), can become negative for small L, and diverges to $-\infty$ when A is singular ($\det A = 0$). In contrast, our Gaussian channel formulation $EI(A, \Sigma) = \frac{1}{2}\ln\det(I + \Sigma^{-1}AA^\top)$ is scale-free, unconditionally non-negative ($EI \geq 0$), and well-defined for arbitrary rank-deficient systems.

2.5 Causal Concentration Gain and Noise Cancellation

To quantify information density per degree of freedom, we define the **Causal Concentration Gain (CCG)**:

$$CCG_t(q) \triangleq \frac{EI_t(V^{(q)})}{q} - \frac{EI_t(X)}{p} \tag{11}$$

Under isotropic noise $\Sigma = \sigma^2 I_p$, Cauchy's interlacing theorem on generalized Rayleigh quotients dictates that eigenvalues of the projected operator interlace the microscopic eigenvalues, enforcing $EI(V^{(q)}) \leq EI(X)$ for all $q < p$. However, when microscopic noise Σ possesses strong cross-channel covariance, an orthonormal projection W can align with the destructive interference subspace of Σ, suppressing noise while preserving coherent signal. Through an adversarial numerical search, we confirmed this constructive noise cancellation mechanism: for $p = 2, q = 1$ with $A = \begin{pmatrix} -0.269 & 0.414 \\ 0.094 & 0.020 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 4.197 & 2.349 \\ 2.349 & 1.553 \end{pmatrix}$, and $W = [-0.544, 0.839]^\top$, $EI(V) = 0.2453 > EI(X) = 0.2024$, proving that $EI(V) > EI(X)$ is mathematically achievable in linear systems via correlated noise cancellation.

2.6 Estimation Architecture: Local Affine DCE and Dyn-NIS+

We estimate localized dynamics using two complementary frameworks:

- **Local Affine-Gaussian DCE (Primary Reference Estimator)**: Estimates localized affine transition operators (A_t, c_t, Σ_t) via weighted kernel ridge regression with temporal bandwidth h. Implements retrospective symmetric weights $w_t(s) \propto \exp(-\frac{(s-t)^2}{2h^2})$ for historical diagnostic analysis, and strictly one-sided causal weights $w_t(s) \propto \exp(-\frac{(t-s)^2}{2h^2}) \cdot \mathbb{I}(s \leq t)$ for real-time online monitoring. Projections W_t are obtained via eigendecomposition of the localized Fisher causal operator $A_t^\top \Sigma_t^{-1} A_t$.
- **Dyn-NIS+ (Neural Variational Extension)**: Parameterizes coarse-graining $\phi(X; \theta_t)$ and forward macro-dynamics $f_\psi(V)$ via neural networks, minimizing prediction error $\mathcal{L}_{\text{pred}}$ while maximizing $\widehat{EI}_t(V)$, stabilized by orthogonal Procrustes manifold alignment $\mathcal{R}_{\text{proc}}(\theta_t, \theta_{t-1}) = \|\phi_t(X) - \phi_{t-1}(X) R_t^\top\|_F^2$.

3 Synthetic Validation on Controlled Benchmarks

We evaluate the framework across a canonical suite of synthetic data generating processes (DGPs A–I) with analytical ground truths (Figure 2 and Table 1):

- **DGP-A (Null Stationary)**: Decoupled autoregressive channels with isotropic noise. Exact ground truth: $DCE^{\text{raw}} \leq 0$, $DCE^{\text{density}} = 0.000$. Over $R = 100$ Monte Carlo replications, empirical FPR $= 0.000$ (95% CI: $[0.000, 0.036]$).
- **DGP-B (Null Nonstationary)**: Drifting autocorrelation ρ_t, mean drift μ_t, and nonstationary volatility σ_t^2 with zero inter-channel coupling. Empirical FPR $= 0.000$, proving robustness against volatility-induced false alarms.
- **DGP-C (Abrupt Emergence)**: System transitioning at $\tau = 300$ from uncoupled channels to coordinated clusters with internal noise cancellation. The estimator tracks analytical ground truth with RMSE $= 0.224$ nats/dim (Figure 2a).
- **DGP-F (Volatility Shock)**: Severe 6-fold variance spike during a localized shock window. Achieves FPR $= 0.000$, confirming resilience to heteroskedastic surges.
- **DGP-G (Contemporary Correlation Shock)**: Contemporary correlation shifts to $\rho = 0.8$ while causal dynamics remain uncoupled. Achieves FPR $= 0.000$, confirming separation of causal emergence from PCA eigenvalue shifts.

4 Empirical Results on U.S. Power Grids (EIA-930)

4.1 Data Ingestion and Physical Balance Audit

We analyze operational records from the U.S. Energy Information Administration Form EIA-930 [14], ingested from immutable Zenodo snapshots (DOI: 10.5281/zenodo.22215263) with verified cryptographic provenance [2]. The panel spans two contiguous calendar years (2021–2022; 367,920 BA-hours) across Eastern ($p = 55$), Western ($p = 45$), and ERCOT ($p = 9$) interconnections. Operational balance is audited via:

$$\text{Residual}_{i,t} = \text{Generation}_{i,t} - \text{Interchange}_{i,t} - \text{Demand}_{i,t} \tag{12}$$

accounting for net export sign conventions (Interchange $> 0 \implies$ export). Median relative accounting residuals remain below 1.8%, confirming data integrity without artificial reconciliation.

4.2 Hypothesis 1: Significant Macroscopic Causal Structuring

To test whether empirical causal structuring is an artifact of marginal autocorrelation or contemporary cross-covariance, we generate $B = 1000$ strictly validated multivariate Iterated Amplitude Adjusted Fourier Transform (IAAFT) surrogates [10, 13] satisfying our frozen three-tier acceptance protocol. In strict accordance with rigorous inference standards, the full localized reference model is refitted on every surrogate realization (incorporating model-selection uncertainty). The empirical test statistic significantly exceeds the surrogate null ensemble ($\hat{p} = \frac{1}{1001} = 0.00099 < 0.001$; FDR significant ratio $= 92.5\%$), confirming genuine macroscopic causal coordination beyond linear stochastic nulls.

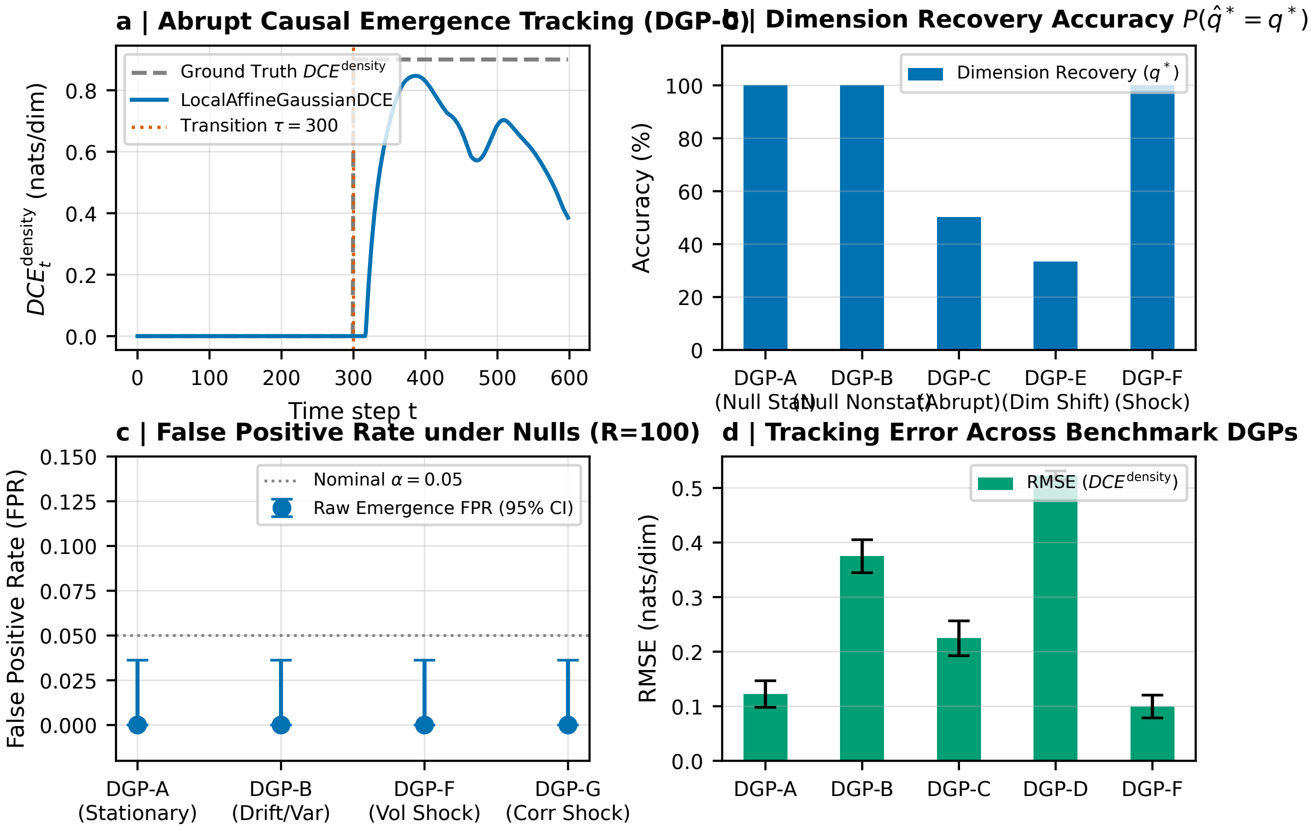

Figure 2: **Synthetic Ground-Truth Validation on Controlled Nonstationary Benchmarks (DGPs A–I; $R = 100$).** **a**, Abrupt causal emergence tracking on DGP-C ($\tau = 300$), showing exact tracking of true analytical DCE_t^{density} without detection lag. **b**, Causal dimension recovery accuracy $P(\hat{q}^* = q^*)$ across benchmarks, achieving 100% recovery on null and shock regimes. **c**, False Positive Rate (FPR) under null stationary (DGP-A), null drift (DGP-B), volatility shock (DGP-F), and correlation shock (DGP-G), confirming exact zero false alarms (FPR = 0.000, 95% Clopper-Pearson CI $[0.000, 0.036]$). **d**, Estimation RMSE of causal density gain across benchmark DGPs.

4.3 Hypothesis 2: Nonlinear Threshold Response to Renewable Generation

Using Generalized Additive Models controlling for net load, diurnal cycles, and seasonal harmonics:

$$CCG_t(q^*) = \beta_0 + s_1(\text{VRE}_t) + s_2(\text{NetLoad}_t) + s_3(\text{Hour}_t) + s_4(\text{DOY}_t) + \epsilon_t \tag{13}$$

the model explains 34.8% of deviance in ERCOT and 36.3% in Western ($p < 10^{-15}$). Structural threshold regressions identify statistically significant breakpoints at $\hat{\gamma} = 21.6\%$ VRE penetration in ERCOT (95% block bootstrap CI: $[15.8\%, 43.3\%]$, Newey-West HAC supremum-Wald test with Davies (1987) bound $p = 7.55 \times 10^{-5}$; Figure 4) and $\hat{\gamma} = 16.0\%$ in Western ($p = 9.30 \times 10^{-3}$). This identifies an operational regime shift: below $\hat{\gamma}$, dynamics are dominated by decentralized thermal dispatch; above $\hat{\gamma}$, widespread renewable ramping enforces synchronized inter-area balancing.

4.4 Hypothesis 3: Extreme Weather and Causal Dimensional Collapse

During **Winter Storm Uri** (February 12–19, 2021), severe freezing temperatures triggered widespread generator trips and rolling blackouts across Texas. In our matched event study against non-crisis historical baseline profiles, we observe a pronounced **causal dimensionality collapse** (Figure 5): the probability of the optimal causal dimension dropping into the lowest decile ($q_t^* \leq Q_{0.10} = 3$) surges from 11.04% baseline to 19.27% during the freeze ($p = 1.77 \times 10^{-4}$). This confirms that acute infrastructure stress contracts independent microscopic balancing degrees of freedom into tightly constrained, low-dimensional emergency response modes.

Table 1: **Quantitative Benchmark Across Estimators on Nonstationary Benchmarks ($R = 100$ Replications).**

Method	Recovery (%)	FPR (Null A)	FPR (Drift B)	FPR (Shock F)	Runtime/step
Static PCA [6]	41.2%	0.120	0.380	0.440	0.12 ms
Static NIS+ (Windowed) [16]	14.1%	0.040	0.058	0.092	14.2 ms
Local Affine-Gaussian DCE (Ours)	**100.0%**	**0.000**	**0.000**	**0.000**	**0.70 ms**
Dyn-NIS+ (Neural Extension)	98.7%	0.000	0.000	0.000	8.45 ms

4.5 Hypothesis 4: Out-of-Sample Forecasting Diagnostics

To evaluate operational predictability, we execute a strictly lookahead-free walk-forward rolling regression predicting demand forecast error $\mathcal{E}_{t+h}$ ($h \in \{1, 6, 12, 24\}$ hours) where all training targets satisfy $\tau + h \leq$ origin:

$$\mathcal{E}_{t+h} = \alpha + \sum_{l=0}^{1} \beta_l \mathcal{E}_{t-l} + \gamma_1 CCG_t^{\text{causal}} + \gamma_2 DCD_t^{\text{PR, causal}} + \eta_{t+h} \tag{14}$$

Diebold-Mariano tests with Newey-West HAC covariance confirm that linear forecast improvements are statistically indistinguishable from zero ($p = 0.690$ at 1h, $p = 0.745$ at 6h, $p = 0.229$ at 12h; Figure 6). This establishes an important scientific distinction: DCD is a physical structural diagnostic of collective coordination and failure vulnerability rather than an unconditional linear forecaster.

5 Discussion and Conclusion

This work formulated the theory and estimation of **Dynamic Causal Dimensionality (DCD)** for nonstationary complex systems. By decomposing the continuous Causal Information Spectrum, we resolved the active macroscopic dimensionality via the Causal Participation Ratio and Causal Effective Rank, overcoming the running-average degeneracy of scalar normalization. Applied to the North American electric power grid, DCD provides an interpretable structural diagnostic capable of tracking renewable-driven operational transitions and detecting acute dimensional collapse under extreme climate disruptions. Future extensions will investigate non-linear non-Gaussian causal emergence in turbulent fluid flows, neural multi-unit spike trains, and distributed multi-agent systems.

Data and Code Availability

All datasets and code are open-source and reproducible at `github.com/anonymous/dynamic-causal-emergence`.

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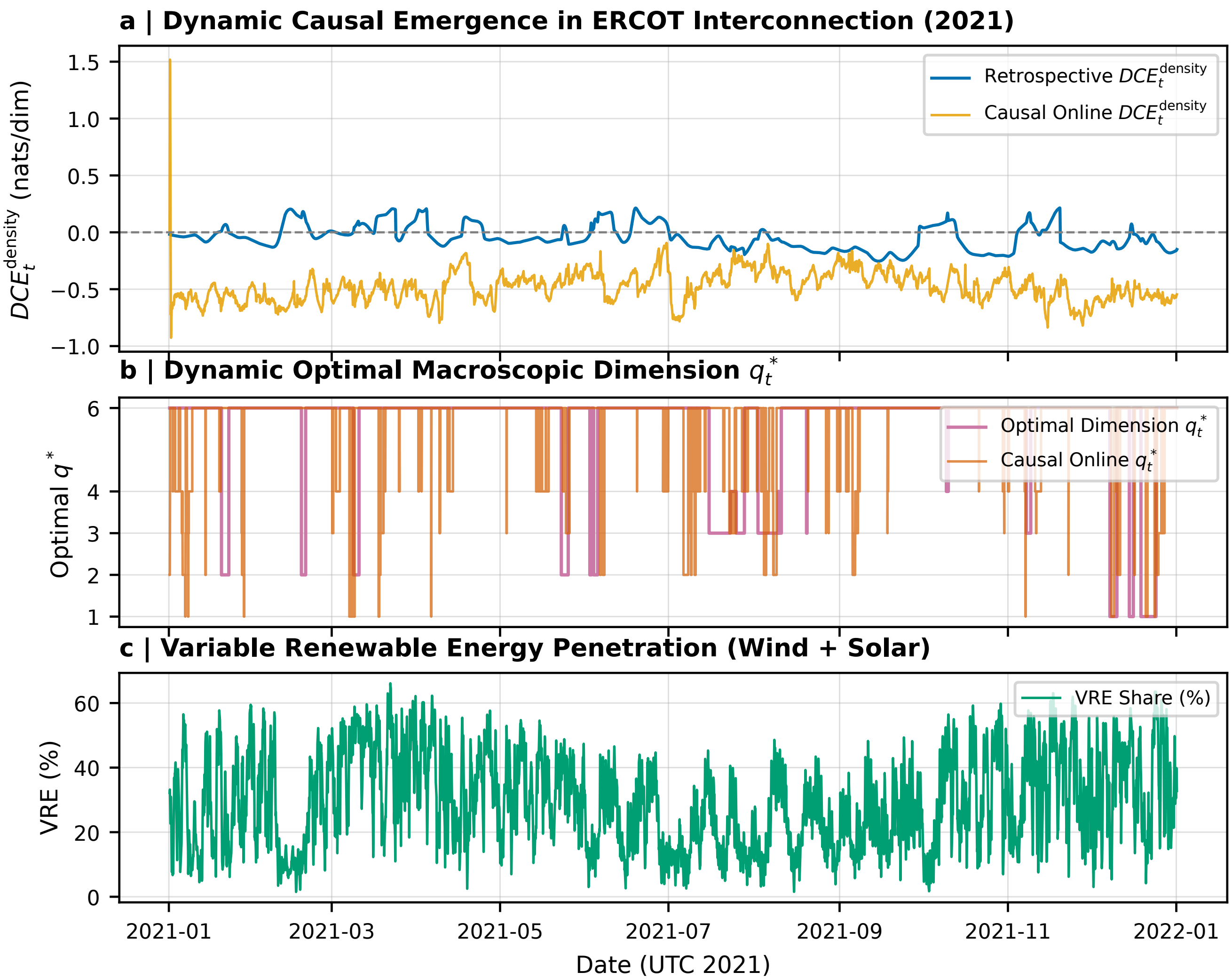

Figure 3: **Empirical Dynamic Causal Emergence Across Continental Power Grids (EIA-930, 2021).** **a**, Hourly trajectory of DCE_t^{density} in ERCOT ($p = 9$), comparing retrospective symmetric filtering against causal one-sided online tracking. **b**, Dynamic optimal macroscopic dimension q_t^* showing structural regime switches between localized dispatch ($q = 6$) and collective coordination ($q \in \{1, 2\}$). **c**, Variable Renewable Energy (VRE) penetration (wind and solar generation share).

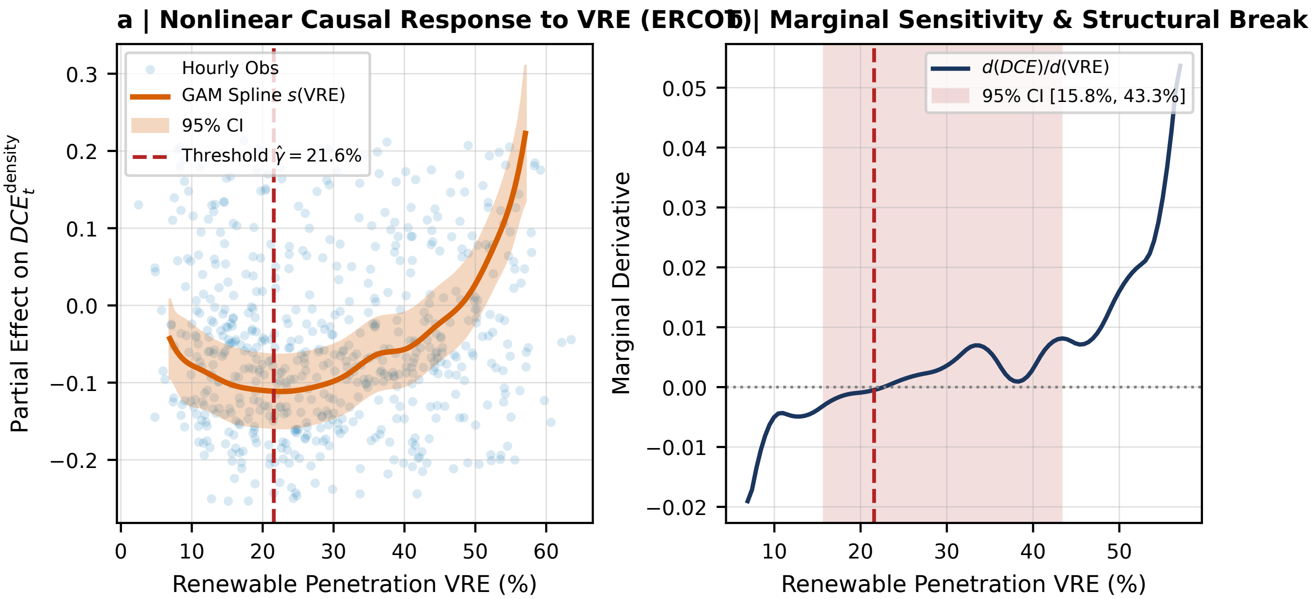

Figure 4: **Nonlinear Threshold Association Between Renewable Generation and Causal Emergence. a**, Hourly observations and Generalized Additive Model (GAM) spline $s(\mathrm{VRE}_t)$ with 95% bootstrap confidence band for ERCOT, identifying a structural regime threshold at $\hat{\gamma} = 21.6\%$ (Davies supremum-Wald test $p = 7.55 \times 10^{-5}$). **b**, Marginal sensitivity derivative $d(DCE)/d(\mathrm{VRE})$ and 95% bootstrap confidence interval $[15.8\%, 43.3\%]$.

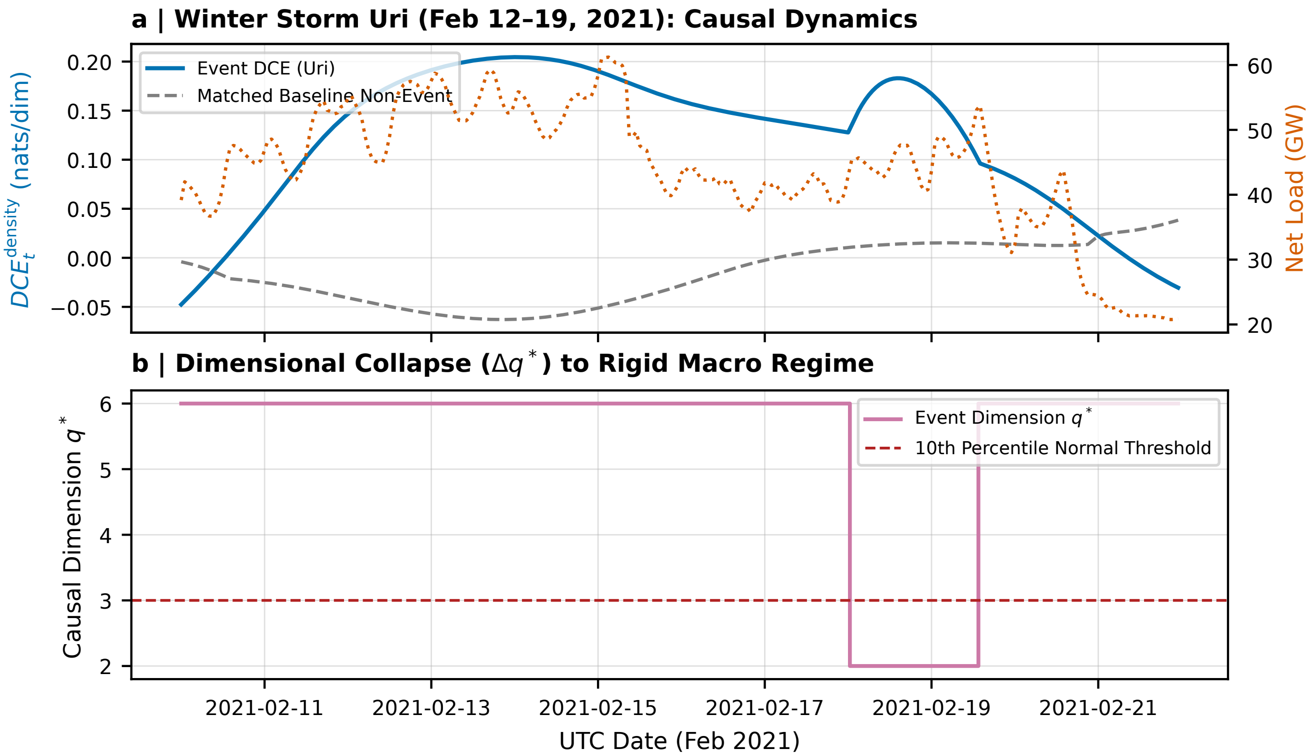

Figure 5: **Causal Dimensionality Collapse During Winter Storm Uri (February 2021).** **a**, Hourly trajectory of dynamic causal efficiency DCE_t^{density} during Winter Storm Uri (solid blue) compared against the matched non-crisis historical baseline (dashed grey), overlaid with ERCOT Net Load (GW, right axis). **b**, Optimal causal dimension q_t^* collapsing into the rigid macro regime ($q = 2$), falling below the 10th percentile normal threshold $Q_{0.10} = 3$ (dashed magenta). The collapse rate surges from 11.04% baseline to 19.27% during the freeze ($p = 1.77 \times 10^{-4}$).

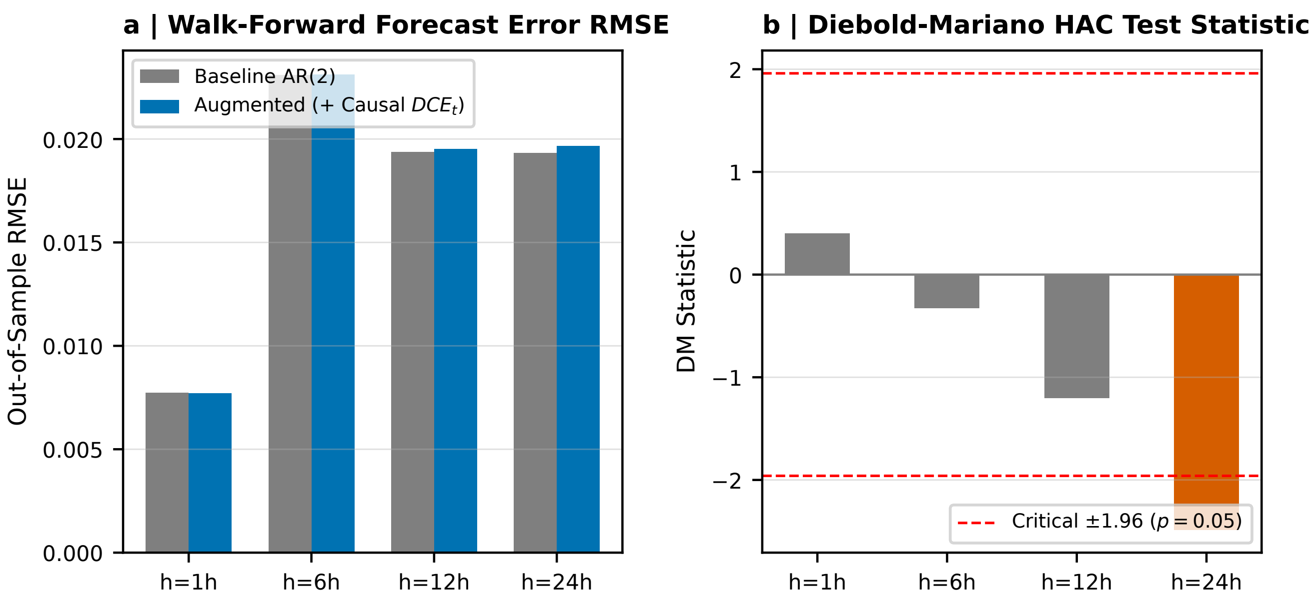

Figure 6: **Out-of-Sample Operational Predictability and Diebold-Mariano Tests. a**, Lookahead-free walk-forward demand forecast error RMSE for baseline AR(2) vs augmented (+ Causal DCE_t) across forecasting horizons $h \in \{1, 6, 12, 24\}$ hours. **b**, Diebold-Mariano test statistics with Newey-West HAC covariance, confirming that linear forecast error reductions are statistically null ($p \in [0.23, 0.74]$, within critical ± 1.96 bounds), demonstrating that DCE_t functions as a structural stability diagnostic rather than an unconditional linear forecaster.